

UNIVERSIDAD DE EL SALVADOR

Facultad Multidisciplinaria De Occidente Departamento
de Ingeniería y Arquitectura Ingeniería en Desarrollo de
Software

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Asignatura: Matemática aplicada

Tema: Guía

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**G
U
I
A**



a) $f(x, y) = 3x + 5y + 2$

$D = \{(x, y) \in \mathbb{R}^2\}$

Graficar

$z = 3x + 5y + 2$

Cortes

Eje z

$x = 0$

$y = 0$

$z = 2$ es el plano cortado eje z $(0, 0, 2)$

Eje x

$y = 0$

$z = 0$

$0 = 3x + 2 \rightarrow x = -\frac{2}{3}$ (corta $-\frac{3}{3}, 0, 0$)

Eje y

$x = 0$

$z = 0$

$0 = 5y + 2 \rightarrow y = -\frac{2}{5}$ (corta y en $0, -\frac{2}{5}, 0$)

Dominio $\mathbb{R}^2$

Graficar

eje z :

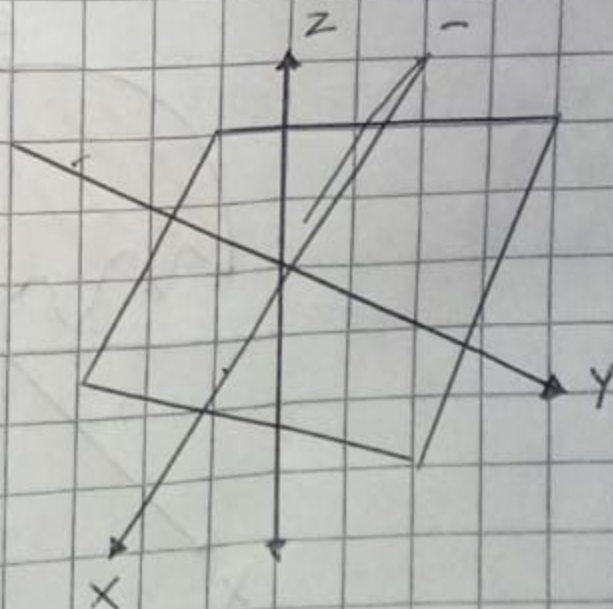
$(0, 0, 2)$

eje x :

$(-\frac{2}{3}, 0, 0)$

eje y :

$(0, -\frac{2}{5}, 0)$



$z = 3x + 5y + 2$

b) $g(x, y) = \sqrt{9 - x^2 - y^2}$

$$9 - x^2 - y^2 \geq 0$$

$$\downarrow$$

$$x^2 + y^2 = 9$$

Domínio = $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\}$

Graficar, se limitan los valores donde " x " $\in (-3, 3)$,
" y " $\in (-3, 3)$

Radio = 3

Centro = $(0, 0)$

Punto vertice $z = \sqrt{9 - 0^2 - 0^2}$
 $= \sqrt{9}$
 $= 3$

Borde del círculo (A)

$$x^2 + y^2 = 9 \rightarrow z = \sqrt{9 - 9}$$

$$= 0$$



o)

$$f(x, y) = \sqrt{y + 4x^2 - 4}$$

$$y + 4x^2 - 4 \geq 0 \rightarrow y \geq 4 - 4x^2$$

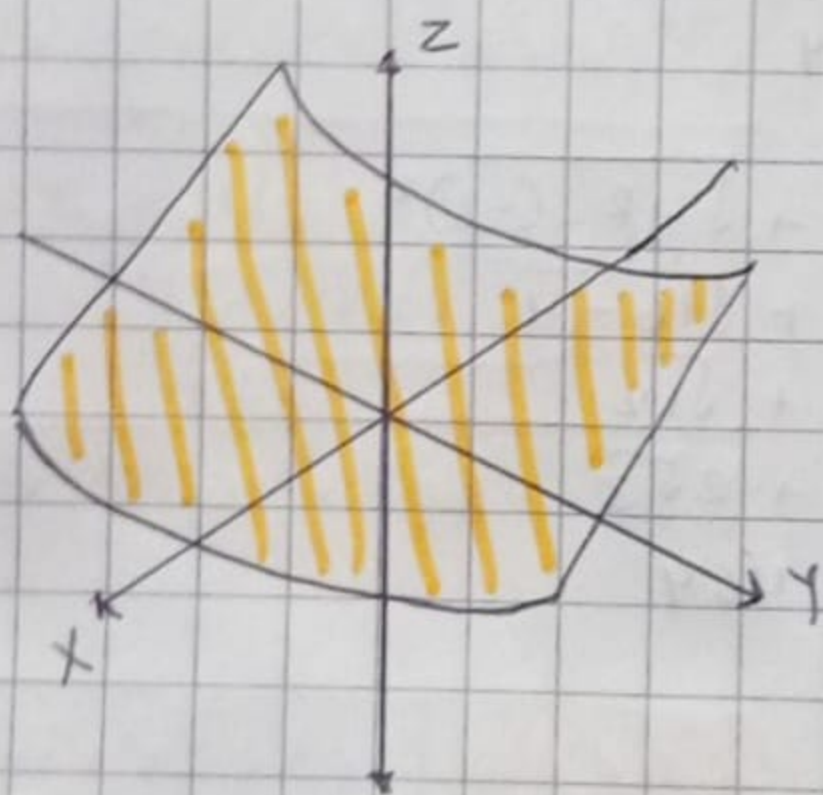
Para el valor de x el valor mínimo de " y " está dado por la parábola $y = 4 - 4x^2$

El dominio se encuentra encima de la parábola

$$\text{Dominio : } D = \{(x, y) \in \mathbb{R}^2 \mid y \geq 4 - 4x^2\}$$

$$\text{Cortes : } (0, 4), z = 0$$

$$(1, 4), z = 2$$



d) $f(x, y) = 4 + \sqrt{x^2 - y^2}$

$$x^2 - y^2 \geq 0 \rightarrow x^2 \geq y^2$$

$$D = \{ (x, y) \in \mathbb{R}^2 \mid x \geq y \}$$

$$\begin{aligned} \bullet f(1, 0) &= 4 + \sqrt{1^2 - 0^2} \\ &= 4 + \sqrt{1 - 0} \\ &= 4 + 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} \bullet f(5, 3) &= 4 + \sqrt{5^2 - 3^2} \\ &= 4 + \sqrt{25 - 9} \\ &= 4 + \sqrt{16} \\ &= 4 + 4 \\ &= 8 \end{aligned}$$

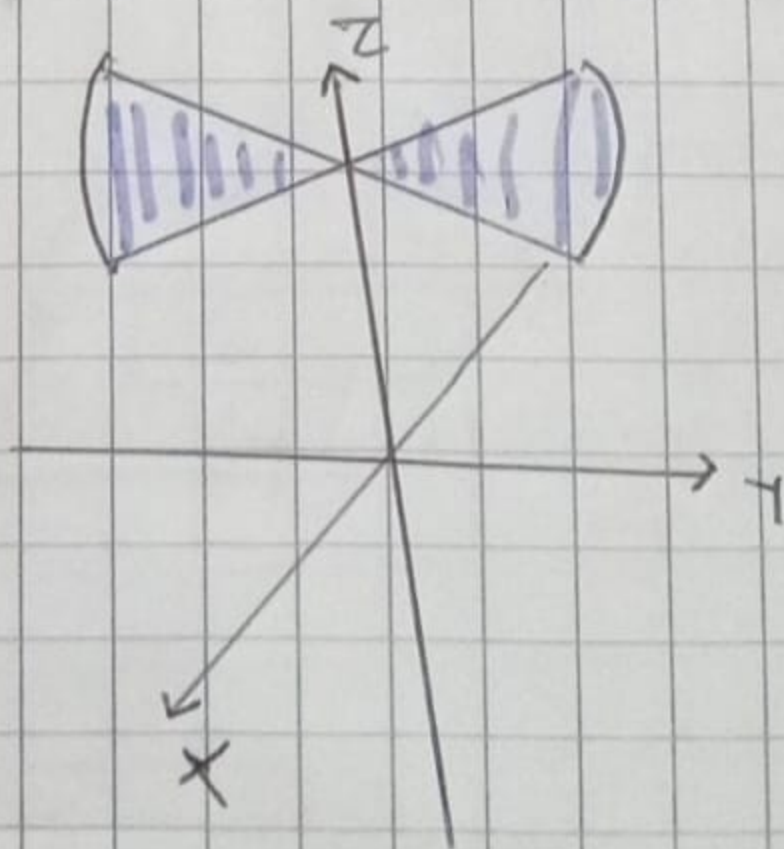
$$\begin{aligned} \bullet f(4, -2) &= 4 + \sqrt{4^2 - (-2)^2} \\ &= 4 + \sqrt{16 - 4} \\ &= 4 + \sqrt{12} \\ &= 4 + 2\sqrt{3} \\ &= 7.464 \end{aligned}$$

Gráfico:

$$(1, 0), z = 5$$

$$(5, 3), z = 8$$

$$(4, -2), z = 7.464$$



e) $f(x, y) = \sqrt{2x^2 + 4xy - x - 2y}$

$$2x^2 + 4xy - x - 2y \geq 0$$

Factorizar

$$2x^2 + 4xy - x - 2y = 2x(x + 2y) - (x + 2y)$$

$$(2x - 1)(x + 2y) \geq 0$$

Resolución de desigualdad

$$2x - 1 = 0$$

|

$$x = 1/2$$

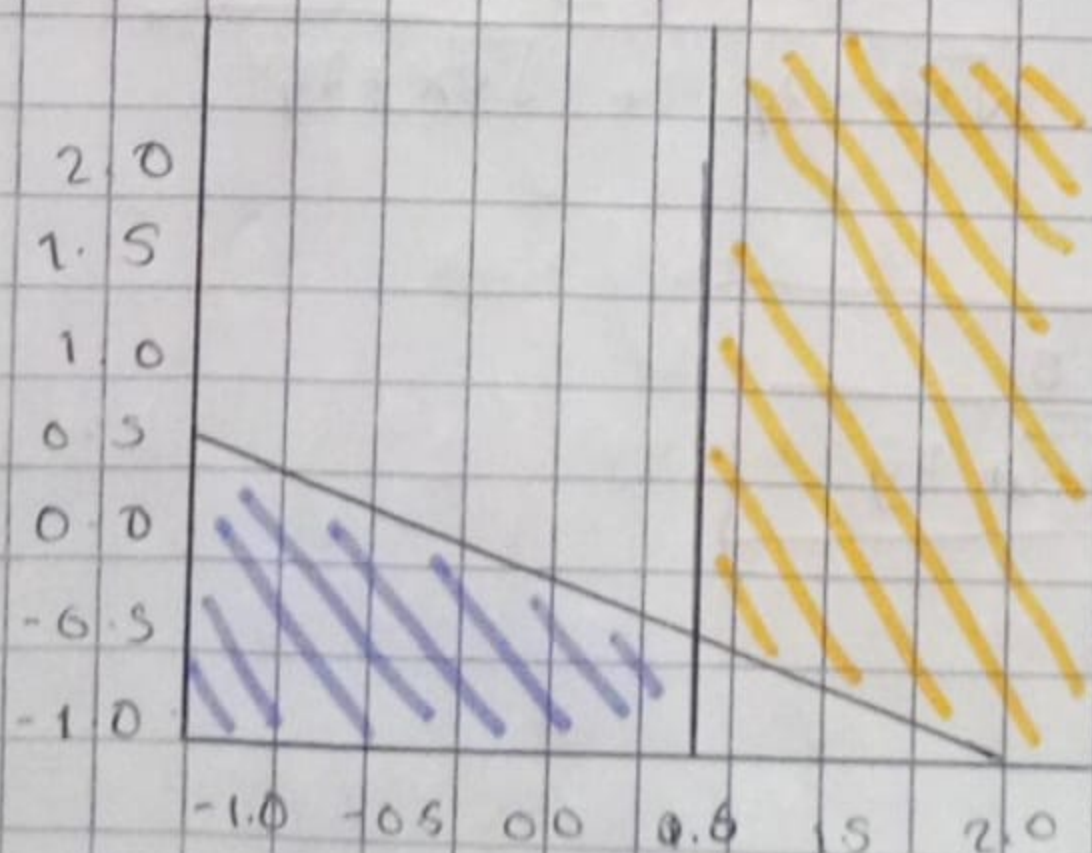
$$x + 2y$$

|

$$y = -x/2$$

Graficar

- Vertical $x = 1/2$
- Ordinada $y = -x/2$
- Sombrar región: $(2x - 1)(x + 2y) \geq 0$



$$(f) \quad f(x, y) = \sqrt{5x+7y}$$

$$f_x(x, y) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$f_x(x, y) = \lim_{\Delta x \rightarrow 0} \frac{\sqrt{5(x + \Delta x) + 7y} - \sqrt{5x + 7y}}{\Delta x}$$

$$f_x(x, y) = \lim_{\Delta x \rightarrow 0} \frac{[\sqrt{5(x + \Delta x) + 7y} - \sqrt{5x + 7y}][\sqrt{5(x + \Delta x) + 7y} + \sqrt{5x + 7y}]}{\Delta x [\sqrt{5(x + \Delta x) + 7y} + \sqrt{5x + 7y}]}$$

$$\Delta x [\sqrt{5(x + \Delta x) + 7y} + \sqrt{5x + 7y}]$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(5(x + \Delta x) + 7y) - (5x + 7y)}{\Delta x [\sqrt{5(x + \Delta x) + 7y} + \sqrt{5x + 7y}]}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{5\Delta x}{[\sqrt{5(x + \Delta x) + 7y} + \sqrt{5x + 7y}]}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{5}{\sqrt{5x + 7y} + \sqrt{5x + 7y}}$$

$$= \boxed{\frac{5}{2\sqrt{5x + 7y}}}$$

$$f_y(x, y) = \lim_{\Delta y \rightarrow 0} \frac{\sqrt{sx+7}(y+\Delta y) - \sqrt{sx+7}y}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{[\sqrt{sx+7}(y+\Delta y) - \sqrt{sx+7}y][\sqrt{sx+7}(y+\Delta y) + \sqrt{sx+7}y]}{\Delta y [\sqrt{sx+7}(y+\Delta y) + \sqrt{sx+7}y]}$$

$$\Rightarrow \frac{[sx+7(y+\Delta y) - (sx+7y)]}{\Delta y [\sqrt{sx+7}(y+\Delta y) + \sqrt{sx+7}y]}$$

$$\Rightarrow \frac{7\Delta y}{\Delta y [\sqrt{sx+7}(y+\Delta y) + \sqrt{sx+7}y]}$$

$$R/I \quad f_y(x, y) = \frac{7}{2\sqrt{sx+7}y} \quad y(f_y)$$

$$f_x(x, y) = \frac{s}{2\sqrt{sx+7}y} \quad x(f_x)$$

$$g) \int_0^2 \int_0^1 (1+2x+2y) dy dx$$

$$0 \leq x \leq 2$$

$$0 \leq y \leq 1$$

$$\begin{aligned} \int_0^2 (2+2x) dx &= 2x + x^2 \Big|_0^2 \\ &= (2(2) + 2^2) - (2(0) + 0^2) \\ &= 4 + 4 - 0 \\ &= 8 \end{aligned}$$

$$\begin{aligned} \int_0^1 (1+2x+2y) dy &= y + 2xy + y^2 \Big|_0^1 \\ &= (1+2x(1)+(1)^2) - (0+2x(0)+(0)^2) \\ &= 1 + 2x + 1 - 0 \\ &= 2 + 2x \end{aligned}$$

$$h) \iint (1+6xy^2) dx dy$$

$$\begin{aligned} \int (1+6xy^2) dx &= \int 1 dx + \int 6xy^2 dx \\ &= x + 6 \int x dx \end{aligned}$$

$$= x + 6y^2 \left(\frac{x^2}{2} \right) = x + 3y^2 x^2$$

$$\begin{aligned} \int (x + 3y^2 x^2) dy &= \int x dy + \int 3y^2 x^2 dy \\ &= xy + 3x^2 \left(\frac{y^3}{3} \right) \\ &= xy + x^2 y^3 \\ &= xy + x^2 y^3 \end{aligned}$$